AFONSO PINTO CARDOSO RAFAEL

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EDUCATION

Masters in Computer Science, Faculty of Sciences of the University of Lisbon
(paused for full-time research position)
- Relevant Coursework: Compilation Techniques, Domain Specific Languages, Quantum Computing.

B.S. in Geospatial Engineering, Faculty of Sciences of the University of Lisbon 2016 - 2020

Drumming, International Drum Academy 2016 - 2020

SKILLS

Programming languages	C, C++, Fortran, LLVM, Haskell, Lisp, Typescript, Python
Technical Skills	Compiler design and implementation, Web Development, Data processing
Soft Skills	Enjoy speaking with people, Enjoy presentations and public speaking
Communication languages	Português (nativo), Español (avanzado, uso profesional), Català (Bàsic 1) English (advanced, professional use)

EXPERIENCE

Compiler Researcher/Engineer (RE2) May 2026 - Now
Barcelona Supercomputing Center *Barcelona, ES*

- Work on openMP target offloading in LLVM/Clang.
- Organize the compiler's team book club (reading SSA-based Compiler Design)

Compiler Researcher/Engineer (RE1) Nov 2023 - Now
Barcelona Supercomputing Center *Barcelona, ES*

- Work on loop fusion within the LLVM infrastructure.
- Benchmark and test the loop fusion pass.
- Contribute with issues and patches in LLVM (example: tidy warnings, duplicated statistics, DT bug)

Web Developer July 2023 - Nov 2023
Digital MZN *Lisbon, PT*

- Developed features for the product website using React Native and Typescript.
- Help with setting up python environments of other members of the company.

Software Developer (Freelancing) April 2023 - May 2023
Bynd *Lisbon, PT*

- Build a web crawler in python to collect data on companies to be displayed on a dashboard.
- Using selenium as a dependency to allow interaction between the crawler and the web page.

Researcher (Intern) September 2022 - March 2023
IMDEA Software *Madrid, ES*

- Build relational benchmarks for Liquid Haskell, a compiler plugin that extends Haskell's type system with refinement types.
- Wrote a paper section describing the benchmarks.
- Contributed with solving a bug in liquid-fixpoint (Pull Request)

Research Fellowship January 2022 - July 2022
LASIGE *Lisbon, PT*

- Freest is a functional programming language that uses session types for safer communication.
- Developed a compiler for the Freest language, targeting JavaScript (Built with Haskell).
- Built a test suite for the compiler.
- Best Poster - 1st, LASIGE Workshop'22.

Summer with science (Research Fellowship) August 2021
LASIGE *Lisbon, PT*

- Built programs with Freest that later got into the testing suite.
- Found bugs and reported them during development.

Software Developer (Intern) 2019 - 2020
Think Future *Lisbon, PT*

- Built a system to manage aerial information for control of drone flight (Python).

Data Science (Intern) August 2019 - September 2019
ITC Faculty of Geo-information Science and Earth Observation *Ensched, NL*

- Wrote data processing code using Dask, an open-source Python library for parallel computing.
- Lectured practical classes on the Institute.
- Processed geographical information for decision making.

OUTSIDE OF JOB

Discord Admin/Event Manager 2020 - Present
DevPT *Lisbon, PT*

- Organize and maintain the devPT Book club
- Participated in the devPT talks as Manager.
- Participated in the devPT as Speaker (Eu uso Emacs, roast me.)
- Moderate and Help make decisions on the Discord platform.
- Help manage social media.

Open Source Contributor

- Fix warnings in sclang mode (pull request).

MUSIC

Drummer 2019 - 2023
AZMA *Lisbon, PT*

- Play drums and help compose the songs.
- Help manage the social accounts and project contacts.
- Participated on the recording of songs like: Akadom

Guitarist 2022 - 2023
Split Objects *Lisbon, PT*

- Support guitarist for the band.
- Help with gigs.

Drummer 2017 - 2019
PodPop *Lisbon, PT*

- Played drums and helped compose music.
- Helped record the video clips.

- Participated on the recording of songs like: Is It Love Or Sacrifice, BlackOut, Spoon Of The Gods.

Guitarist 2016
Why Knot *Lisbon, PT*

- Played guitar and helped compose music.
- Participated on the recording of songs like: Nothing Concrete.